16.1 In Examples 16.2 and 16.4, we presented the linear probability and probit model estimates using an example of transportation choice. The logit model for the same example is $P(AUTO = 1) = \Lambda\left(\gamma_1 + \gamma_2 DTIME\right)$, where $\Lambda(\bullet)$ is the logistic *cdf* in equation (16.7). The logit model parameter estimates and their standard errors are

$$\begin{array}{rl} \tilde{\gamma}_1 + \tilde{\gamma}_2 DTIME = & -0.2376 + 0.5311 DTIME \\ \text{(se)} \qquad\qquad & \;\;(0.7505)\;\;\;(0.2064) \end{array}$$

a. Calculate the estimated probability that a person will choose automobile transportation given that $DTIME = 1$.

b. Using the probit model results in Example 16.4, calculate the estimated probability that a person will choose automobile transportation given that $DTIME = 1$. How does this result compare to the logit estimate? [*Hint*: Recall that Statistical Table 1 gives cumulative probabilities for the standard normal distribution.]

c. Using the logit model results, compute the estimated marginal effect of an increase in travel time of 10 minutes for an individual whose travel time is currently 30 minutes longer by bus (public transportation). Using the linear probability model results, compute the same marginal effect estimate. How do they compare?

d. Using the logit model results, compute the estimated marginal effect of a decrease in travel time of 10 minutes for an individual whose travel time is currently 50 minutes longer by driving. Using the probit results, compute the same marginal effect estimate. How do they compare?

a.

DTIME=1

$-0.2376+0.5311\times1=0.2935$

$\Lambda(0.2935)=\dfrac{1}{1+e^{-0.2935}}=0.5279$

DTIME=1, probability $=0.5729$

b.

$$\tilde{\beta}_1+\tilde{\beta}_2 DTIME=-0.0644+0.3000DTIME$$
$$\text{(se)}\qquad\qquad(0.3992)\quad(0.1029)$$

DTIME=1, $\Rightarrow -0.0644+0.3\times1=0.2356$

$\Phi(0.2356)=0.5931$

DTIME=1, probability $=0.5931$

Probit model 所估出的机率较 Logit model 略高一点.

c. Logit: ME 变动

$ME=\dfrac{\partial P}{\partial DTIME}=\lambda(z)\times\tilde{\beta}_2$

DTIME $=3$, $z=-0.2376+0.5311\times3=1.3557$

$ME=\dfrac{e^{-1.3557}}{(1+e^{-1.3557})^2}\times0.5311=0.0865$

在 DTIME=5時，公車車程每增加 10 min，選擇開車的机率提升 0.0865

LPM： ME固定

$ME=\beta_2=0.0703$

在 LPM 下，不論原本車程差距多少，公車車程每增加 10 min

選擇開車的机率提升 0.0703

d. DTIME=-5

Logit： ∵厚尾 ∴ME較小

$z=-0.2376+0.5311\times(-5)=-2.8931$

$$ME=\frac{e^{2.8931}}{(1+e^{2.8931})^2}\times 0.5311=0.0264$$

$\Delta DTIME=-1 \Rightarrow \Delta P=0.0264\times(-1)=-0.0264$

在 Logit 模型下，當事人原本就不太可能開車，若公車時間再相對減少 10 分鐘，選擇開車的機率會再下降約 0.0264

Probit：

$z=-0.0644+0.3(-5)=-1.5644$

$ME=\phi(-1.5644)\times 0.3=0.1173\times 0.3=0.0352$

$\Delta P=0.0352\times(-1)=-0.0352$

公車時間再相對減少 10 分鐘，選擇開車的機率會下降約 0.0352

16.4 In Example 16.3, we illustrate the calculation of the likelihood function for the probit model in a small example. In this exercise, we will repeat that example using logit instead of probit. The logit model for the same example is $P(y = 1) = \Lambda(\gamma_1 + \gamma_2 x)$, where $\Lambda(\bullet)$ is the logistic *cdf* in equation (16.7). The maximum likelihood estimates of the parameters are $\tilde{\gamma}_1 + \tilde{\gamma}_2 x = -1.836 + 3.021x$. The maximized value of the log-likelihood function is -1.612.

a. Calculate the probability that $y = 1$ if $x = 1.5$, given the values of the maximum likelihood estimates.

b. Using the threshold 0.5 and the result in part (a), predict the value of y if $x = 1.5$, the first observation, given the values of the maximum likelihood estimates. Compare your prediction to the actual outcome $y = 1$ in the first observation.

c. Calculate the value of the likelihood function, illustrated in equation (16.14) but substituting equation (16.17) in place of $\Phi(\bullet)$ and using the given $N = 3$ data pairs, if the parameter values are $\gamma_1 = -1$ and $\gamma_2 = 2$. Compare this value to the value of the likelihood function evaluated at the maximum likelihood estimates. Which is larger?

d. For the logit model, the value of the likelihood function (16.14), with $\Lambda(\bullet)$ in place of $\Phi(\bullet)$, will always be between zero and one. True or false? Explain.

e. For the logit model, the value of the log-likelihood function (16.15), with $\Lambda(\bullet)$ in place of $\Phi(\bullet)$, will always be negative. True or false? Explain.

a.

$x=1.5$. $z=-1.836+3.021\times1.5=2.6955$

$$P(y=1|x=1.5)=\frac{1}{1+e^{-2.6955}}=0.9368$$

b. $P(y=1|x)>0.5$ 預測 $\hat{y}=1$

$P(y=1|x)\leq0.5$ 〃 $\hat{y}=0$

by (a). $P(y=1|x=1.5)=0.9368>0.5$ ∴ 預測 $\hat{y}=1$

實際結果和預測結果是一致的，表示模型的預測正確

c.

① $x_1=1.5$ $y_1=1$

$z_1=-1+2\times1.5=2$

$$P(y_1=1)=\frac{1}{1+e^{-2}}=0.8808$$

② $x_2=0.6$, $y_2=1$

$z_2=-1+2\times0.6=0.2$

$$P(y_2=1)=\frac{1}{1+e^{-0.2}}=0.5498$$

③ $\underline{x_3 = 0.7,\ y_3 = 0}$

$z_3 = -1 + 2\times 0.7 = 0.4$

$P(y_3 = 0) = \dfrac{1}{1+e^{0.4}} = 0.4013$

$L(Y_1 = 1, Y_2 = 2) = 0.8808 \times 0.5498 \times 0.4013 = 0.1944$

$\ln(L_{MLE}) = -1.612$

$L_{MLE} = e^{-1.612} = \boxed{0.1995}$ → 大

d. True.

∵ $\Lambda(z) \in (0,1)$

∴ 每一個介於 $(0,1)$ 的數相乘还是会介於 $0\sim1$ 之間.

e. True.

$0 < L < 1$，取 $\ln$

$\Rightarrow \ln L < \ln 1 = 0$

∴ 对数概似函数值恒為負.